

RESPONSE MODEL DATABASE SCHEMA

This data will be used to build our response models.
At the center of this structure are our response and non response events.
The events are surrounded by dimensions that explain why these events happened.

- * Fact Tables are the 'events' that make up a response / non response
- * Dimension Tables describe these events
- * Dimensions attributes should capture (as best we can) all the factors that lead to a respond or non response.

Transaction Fact

Transactions are positive actions that (when done in reaction to a mail piece) are considered a response. Mailed fact only captures transactions that relate to a piece of mail + time frame. Transaction Fact gives us broader insight:

- The history of a persons transactions, stores interchangeable. How many appointments before a purchase
- People who respond outside the matchback window or went to a different client/location.
- False Negatives (the model only learns from its True/False Positives)
- are our responders specific in any way or a random sample of transactions

dim_transaction_type
transaction_type_id
industry_id
transaction_type
transaction_subtype

fact_transaction
transaction_id
location_id
person_id
company_id
transaction_date_id
transaction_type_id
product_service_id
diagnosis_id
contract_id
purchase_value
purchase_quantity
unit_price
source_id
transaction_status
customer_id

dim_product_service
product_service_id
industry_id
industry_label
product_service_type
product_service_subtype
product_service_label
product_brand_id
product_brand_label

dim_product_brand
product_brand_id
industry_id
product_brand_label
manufacturer_id

dim_diagnosis
diagnosis_id
industry_id

dim_contract
contract_id
contract_type

WHO'S BUYING WHAT?

TO WHOM ARE WE SELLING?

WHO IS SELLING?

WHO DID WE TARGET?

Location:

A location is the physical location from where a company sells goods or services.
A location is defined by its owner physical address.
There can never be two locations at the same physical address.
Every location is owned by one and only one company.
A location never changes address, though it may change ownership (company).

Company:

A company is a business entity which sells goods or services.
Companies own locations from which they dispense or provide their goods or services.
Many companies are subsidiaries to a parent company though not all are. Companies can only belong to 1 parent company.
Companies may use to own branding or to buy to a franchise that takes on its parent company brand.

Company type:

A parent company is a company that owns or has a controlling interest in another company or companies.
Parent companies often have a large influence over their subsidiary operations, although they may allow them to perform their operations with little interference.
Parent company type describe the relationship a subsidiary company has with its parent company
It can be an affiliate network, a franchise, Vertical integration, Horizontal integration

dim_industry
industry_id
industry_label
location_id
customer_referral_id
customer_referral_label

dim_manufacturer
manufacturer_id
industry_id
manufacturer_label

dim_parent_company
parent_company_id
parent_company_label
parent_company_type_id
parent_company_type
industry_id
manufacturer_id

dim_company
company_id
industry_id
parent_company_id
parent_company_label
company_type_id
company_type
company_label
franchise_owner

dim_source
source_id
source

dim_location_lookup
lookup_id
company_id
location_id
source_id
source_company_id
source_location_id

dim_location
location_id
address_id
company_id

dim_date
date_id
date
year
day_of_week
day_of_week_short_name
day_of_week_name
is_weekday
is_weekend
day_of_month
week_of_month
month_number
month_short_name
month_name
quarter_number
quarter_name
quarter_number_retail445
quarter_number_retail454
holiday_name
is_holiday

dim_address
address_id
carrier_route_id
address_1
address_2
city
state
zip_code
latitude
longitude

dim_customer
customer_id
location_id
person_id
address_id
source_id
source_create_date

dim_insurance
insurance_id
customer_id
insurance_type
benefit_manager_label
insurance_label
manufacturer_id
manufacturer_label
source
Type

dim_person
person_id
household_id
porch_id
new_mover_id
address_id
first_name
last_name
suffix

dim_new_movers
new_mover_id
person_id
move_date_id
old_address_id
new_address_id

dim_carrier_route
carrier_route_id
carrier_route_code
carrier_route_zip
population_density
business_density
total_area_sq_miles
total_address_count
residential_address_count
business_address_count
residential_address_count_ratio
business_address_count_ratio
residential_population
business_population
total_count
ratio_count
ratio_label
polygons_coordinates

dim_carrier_route_distance
carrier_route_dist_id
carrier_route_id
census_region_id
census_division_id
h3_index_res7
target_carrier_route_id
neighbor_carrier_route_id
neighbor_population
neighbor_address_count
dist_miles

fact_mailed
mail_id
person_id
address_id
location_id
company_id
in_home_date_id
mail_format_id
mail_message_id
score_back_id
piece_price_with_postage
piece_cost
target_type
estimated_postage_cost
currency_id

fact_score
score_back_id
person_id
address_id
model_id
score
score_date_id

dim_currency
currency_id
determination
conversion_rate_to_usd

dim_mail_format
mail_format_id
mail_format_id
mail_message_id
mail_message_id
score_back_id
piece_price_with_postage
piece_cost
target_type
estimated_postage_cost
currency_id

dim_mail_message
mail_message_id
source_id
call_to_action_text
call_to_action_language
call_to_action_font_families
call_to_action_font_sizes
call_to_action_word_count
call_to_action_font_colors
promotional_offer
full_message_text
full_message_language
full_message_font_families
full_message_font_sizes
full_message_word_count
mailer_color_palette